

# Rising Waters

Flood Management System — AI/ML Project

## Project Executable Files

Phase 7: Project Documentation

IBM Cloud

XGBoost 96.55%

Skill Wallet

AI/ML Track

### How to Run the Project

Follow the steps below to set up and run the Rising Waters flood prediction application:

#### Step 1: Clone the Repository

```
git clone https://github.com/Godesivaramakrishna/Flood-Management-System-by-skill-wallet.git
```

#### Step 2: Create Virtual Environment

```
conda create -n flood-env python=3.11 (or python -m venv .venv)
```

```
conda activate flood-env (or .venv\Scripts\activate on Windows)
```

#### Step 3: Install Dependencies

```
python -m pip install -r requirements.txt
```

#### Step 4: Generate Dataset & Train Model (Optional)

```
python model/generate_dataset.py
```

```
python model/train_model.py
```

Note: Pre-trained floods.save is already included in the repository.

#### Step 5: Run Flask Application

```
python app.py
```

#### Step 6: Open in Browser

Navigate to: <http://127.0.0.1:5000>

### Key Files

|                          |                                        |
|--------------------------|----------------------------------------|
| <b>app.py</b>            | Main Flask application — run this file |
| <b>model/floods.save</b> | Deployed XGBoost model (joblib)        |

|                          |                                |
|--------------------------|--------------------------------|
| <b>model/scaler.save</b> | Fitted StandardScaler (joblib) |
| <b>requirements.txt</b>  | All Python dependencies        |
| <b>Dockerfile</b>        | Docker container configuration |
| <b>manifest.yml</b>      | IBM Cloud deployment manifest  |